

TAU-ELICITATION: BENCHMARKING MULTI-TURN ENTITY EXTRACTION IN VOICE AGENTS

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ABSTRACT

Voice agents often need to collect names, addresses, identifiers, dates, and times exactly, yet end-to-end benchmarks obscure where capture fails. We introduce τ -Elicitation, a 200-task voice benchmark spanning 10 entity types, controlled difficulty, caller realisms, and three environments. A text control passes all tasks, but four voice configurations achieve robust exact success on only 14–45%. Agents increase verification for hard and unfamiliar entities and sometimes for incorrect captures, yet only 30–37% of verified errors are repaired. Scaffolded runs have robust Pass³ scores 13–30 percentage points higher and take 21–28 seconds longer per call. Realisms such as spelling variations and restarts do not detectably affect exact success; mispronunciation increases repair effort. These results suggest that strategy selection and successful recovery are key bottlenecks in exact spoken entity collection.

Index Terms— voice agents, spoken dialogue evaluation, multi-turn entity extraction, conversational repair, task-oriented dialogue

1. INTRODUCTION

Voice-agent benchmarks usually score complete tasks [1, 2, 3], yet authentication and key-entity transcription remain common failure points. Existing work does not isolate whether an interactive agent can recognize an uncertain capture, decide when to slow down, repair an error, and store the exact value. Nor does it measure how this capability changes with entity type, difficulty, repeated conditions, or caller effort. This matters because misunderstood named entities impede spoken dialogue, while confirming every attribute can itself frustrate callers [4].

τ -Elicitation tests this capability on short callbacks about missing database fields. Each core task elicits one missing value, isolating the atomic unit of spoken entity collection and establishing a minimum-complexity floor for broader calls; linked multi-entity tasks then test how errors compound. Its default prompt defines the *agent-directed* condition: the record must be exactly correct and letter-by-letter verification is suggested when unsure, but the agent decides when and how to verify. The *scaffolded* condition instead prescribes asking, spelling and read-back, confirming, correcting, and then submitting. Both share consent, never-guess, logging, and one-submission rules, allowing comparison of optional and prescribed verification. Deterministic database comparison scores both conditions without an LLM judge.

We contribute a reproducible 200-task benchmark spanning 10 entity types, controlled difficulty and caller realisms, and multi-

Table 1. Coverage of related benchmarks. Multi: multi-turn; Task: executed tool task; Exact: exact entity score; N : controlled entity count; D: diagnostic or partial coverage.

Work	Multi	Task	Exact	N	Repair
Voice agents [1, 2, 3, 6]	✓	✓	D	–	D
Spoken dialogue [7, 8]	✓	–	✓	–	–
Speech NER [9, 10, 11]	–	–	✓	–	–
τ -Elicitation	✓	✓	✓	✓	✓

entity calls. It measures capture, verification, repair, effort, and robust success across three environments; code and artifacts are available on GitHub [5].

2. RELATED WORK

Existing voice-agent benchmarks— τ -Voice, EVA-Bench, DuplexWorld, and Full-Duplex-Bench-v3—measure end-to-end task performance and interaction quality; several also exercise tool use [1, 2, 3, 6]. Their broad scope reveals failures such as authentication—identified as the dominant bottleneck in τ -Voice and reported separately in EVA-Bench—but does not isolate exact entity capture [1, 2]. τ -Elicitation extends the τ -bench, τ^2 -Bench, and τ -Voice lineage by making controlled voice entity collection the primary task [12, 13, 1].

Dialogue-state tracking spans MultiWOZ, SpokenWOZ, and cross-utterance sub-slot tracking [14, 7, 8]. ASR and spoken-entity work covers natural-speech NER, ASR error propagation, unseen entities in synthesized audio, and industrial missed-entity rates [9, 15, 10, 11]. *Error Detection and Recovery in Spoken Dialogue Systems* studies repair strategies [4]. None combines live tool execution, exact database writes, controlled entity type, difficulty and count, and an agent’s choice to verify. Table 1 summarizes this gap.

3. METHODS

3.1. Task and elicitation policy

Each task is a callback about a record with missing fields. The agent sees the field names, asks the caller for their values, logs its captures, and submits once; the caller reveals a value only after it is requested. The simulated caller is cooperative: it supplies requested values, spells accurately when asked, and confirms or corrects read-backs, but never volunteers information.

3.2. Values, difficulty, and realisms

Ten banks each contain 40 easy and 40 hard values; the benchmark samples 10 of each difficulty per bank. Difficulty is type-relative:

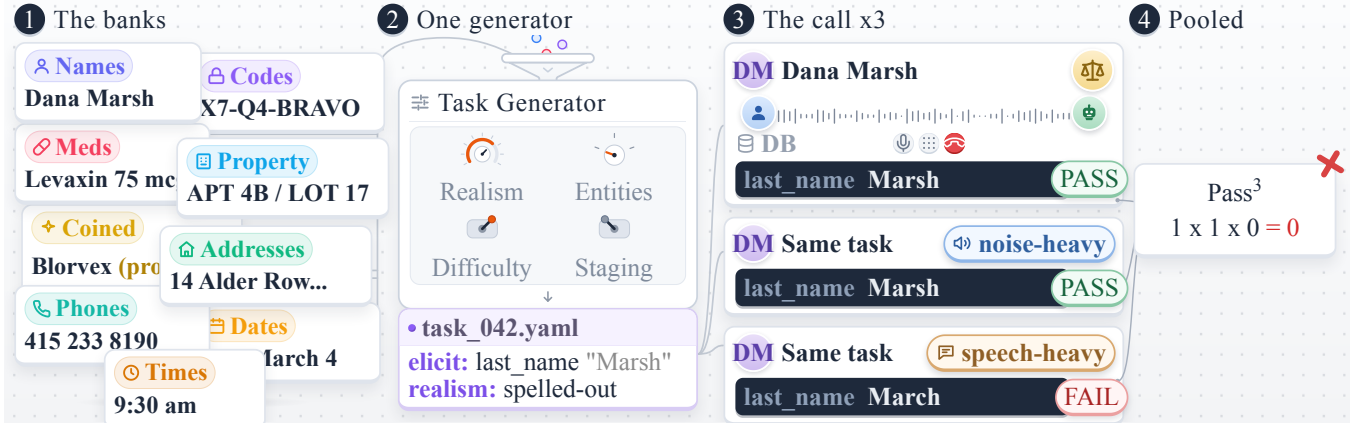


Fig. 1. τ -Elicitation selects values from entity banks, builds a controlled task, and runs it through the τ -Voice interaction loop. Pass³ requires success in all three environment realizations. The scale marks fidelity-judged agent speech.

for example, hard names are rarer, hard codes are longer or confusable, and hard emails contain more symbols. Table 2 reports bank construction and performance.

A seeded catalog adds six caller realisms by appending a scripted instruction to the caller prompt. For example, one spelling-style instruction tells the caller to say “zed” for the letter Z whenever spelling a value. Self-correction supplies a wrong value and immediately repairs it; spelling variation changes how letters or digits are grouped; falter-and-restart spelling introduces a mid-string correction; partial and wrong-field answers require a follow-up; and reviewed mispronunciations alter only the audio. Conditional behaviors occur only when the conversation creates an opportunity, so a spelling correction, for example, cannot fire until spelling begins. Each task stores its generator and catalog versions, seed, selected realism, and expected database state. The generator also emits linked two- and three-field tasks, both as flat calls and as a staged workflow that validates one field before revealing the next.

3.3. Scoring and behavioral measures

Exact success requires database equality after type-specific normalization (e.g., phone-number separators). We report regular-realization success as Pass@1. We define robust Pass³ as exact success on the same task across three prespecified environment realizations: *regular*; *noise-heavy*, with 10 dB background noise, bursts, frame drops, and muffling; and *speech-heavy*, with interruptions, backchannels, and restarts. Unlike a repeated-trial score, Pass³ crosses these prespecified conditions; voices and realism assignments are also redrawn. A 700-call audit placed the 90th-percentile duration at 1.89 minutes, so we use a conservative four-minute limit. Timed-out calls remain in the denominator and score zero.

Post-hoc review found a unit-normalization mismatch. The evaluator now normalizes “milligram(s)” to “mg”.

Behavioral analysis compares first and submitted values with gold after case/punctuation folding. Silent caller tools log every spelling request and read-back; their sum is our measure of verification effort. We compare this effort across noise, entity difficulty and familiarity, and each system’s best- and worst-performing

caller voices.

4. EXPERIMENTS

We evaluate gpt-realtime-2 (minimal and xhigh reasoning), gemini-3.1-flash-live-preview (high), and grok-voice-think-fast-1.0 (provider default) [16, 17, 18]. The caller uses gpt-5.5 (xhigh, temperature zero) [16], with speech synthesized by ElevenLabs eleven.v3 [19].

We compare agent-directed and scaffolded prompting on the same tasks. Each configuration runs all 200 tasks under the regular, noise-heavy, and speech-heavy conditions with each prompt. This gives 600 agent-directed and 600 scaffolded calls per configuration, or 4,800 calls in total. Behavioral, caller-voice, and bank-level analyses use three configurations: GPT xhigh, Gemini high, and Grok.

Main-run confidence intervals come from 10,000 task-level bootstrap samples. When a task is sampled, its three linked conditions stay together. Pairwise system tests also align calls by task. We use a two-sided sign-permutation test with 100,000 permutations and the standard +1 correction. Holm correction is applied separately to the six system comparisons for Pass@1 and Pass³.

Two raters independently review 90 reward-zero calls across providers, labeling failure source and subtype. After discussion and nine explicit adjudications, failure source is resolved for 86 of 90 calls and subtype for 66 of the 81 calls resolved as agent failures. For fidelity validation, gemini-3.1-pro-preview (temperature zero) scores a frozen stratified 60-utterance sample enriched for human flags (30 Gemini and 30 Grok utterances) [17].

5. RESULTS

5.1. Strategy selection and robustness

- Scaffolding trades time for reliability.* Scaffolded runs score 12–36 points higher on Pass@1 and 13–30 points higher on Pass³ (Figure 2), with 21–28 additional seconds per call across the three behavioral configurations. Gains are slightly larger for hard entities (25 versus 21 points).
- A supplied protocol narrows provider gaps.* Grok remains best

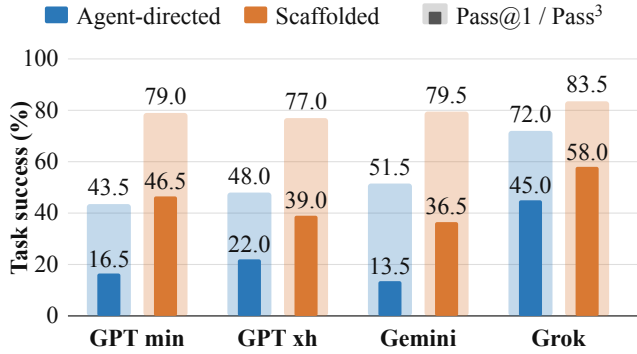


Fig. 2. Task success over 200 single-field tasks for four configurations. Each pair shows agent-directed and scaffolded runs; light full bars show Pass@1, and dark inset bars show robust Pass³.

and significantly exceeds every other agent-directed system on both measures (Holm-adjusted $p < .001$); no other pair is significant. The comparison suggests that explicit verification instructions improve reliability.

- **Robustness exposes instability.** Across systems, Pass³ spans 14–45% agent-directed and 37–58% scaffolded. Rankings can reverse: Gemini beats GPT xhigh on regular Pass@1 (51.5% versus 48.0%), yet trails on Pass³ (13.5% versus 22.0%).
- **Diverse environments expose more than run-to-run variation.** For scaffolded GPT xhigh, three regular runs have similar Pass@1 rates (74.5%, 75.5%, and 77.0%), yet only 103 of 200 tasks pass every repeat (Pass³ = 51.5%): 86 tasks change outcome at least once. Replacing two repeats with noise-heavy and speech-heavy conditions lowers the all-three count to 78 (39.0%). The crossed conditions expose additional instability in this comparison.
- **More reasoning does not consistently improve robustness.** From minimal to xhigh, Pass³ rises from 16.5% to 22.0% agent-directed but falls from 46.5% to 39.0% scaffolded.

5.2. Risk recognition and recovery

In regular agent-directed calls, wrong first captures are logged for 134/200 GPT xhigh fields, 129/199 Gemini fields, and 74/193 Grok fields (67/65/38%); 7/22/14 fields have no capture log. GPT and Gemini verify 66% and 64% of wrong captures versus 49% and 50% of correct ones; Grok verifies nearly everything (93%/96%). The systems likewise spend more effort on hard and unfamiliar entities; GPT xhigh and Grok, but not Gemini, spend more on their lowest-performing caller voice, and only Grok reacts to noise (Figure 3). Verification is selective but imperfect.

Only 30%, 37%, and 32% of verified GPT xhigh, Gemini, and Grok errors end correct. Among 337 initially wrong captures, recovery is 10% without a repair step, 31% with read-back only, 24% with spelling only, and 39% with both. Because agents choose when to verify, these are associations, but the pattern suggests a recovery bottleneck even when verification occurs. Agents can double down on an error or hallucinate a replacement rather than complete the repair.

Caller voice is another blind spot. In agent-directed regular calls, the gap between each system’s best- and worst-performing

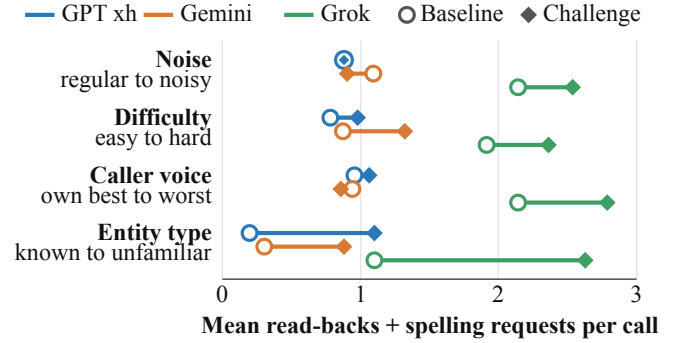


Fig. 3. Mean verification effort (read-backs plus spelling requests). The three systems increase effort for hard and unfamiliar entities. GPT xhigh and Grok, but not Gemini, increase it for their own lowest-Pass@1 caller voice; only Grok increases effort under noise.

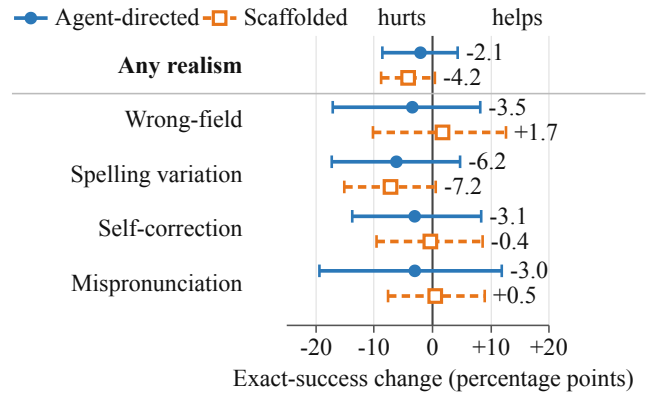


Fig. 4. Caller-realism effects in agent-directed (solid circles) and scaffolded (dashed squares) runs. Bars show task-clustered 95% Hájek intervals versus eligible clean assignments; none survives Holm correction.

voices is 26 points for GPT xhigh, 28 for Gemini, and 12 for Grok. The ordering is not stable: Mildred leads GPT xhigh and Grok, whereas Wei leads Gemini. Only Gemini shows evidence that success differs across all five voices after correction ($p = .040$); no individual Gemini voice pair remains significant, so the result establishes heterogeneity without isolating one contrast. Across systems, Mildred significantly exceeds Mamadou and Arjun, but not Priya or Wei. GPT xhigh and Grok increase verification effort for their weakest voice; Gemini does not.

5.3. Stress tests and diagnostics

Caller realisms show clearer effects on repair cost than success. No assigned realism detectably changes exact success in either arm (Figure 4). Overall Hájek estimates are −2.1 points (95% CI [−8.6, 4.3]) agent-directed and −4.2 ([−8.8, 0.4]) scaffolded. Assigned realisms may remain latent. In agent-directed runs, the mispronunciation success estimate is −3.0 points, while the probability of a spelling request is 24 percentage points higher and calls are 26 seconds longer. These null success tests do not establish absence of an effect.

Performance varies by entity type. Across the regular agent-

Table 2. Value-bank construction and robust agent-directed results. Every bank has 80 values and contributes 20 tasks (10 easy, 10 hard).

Bank	Construction / grounding	Pass ³ (desc)
Dates	Calendar and clock	.60
Phones	Regulator-reserved ranges	.47
Times	Calendar and clock	.43
Person names	SSA + Census [20, 21]	.30
Codes	ISO 3779 VINs; mod-97 IDs	.25
Medications	RxNorm-checked [22]	.23
Properties	Registry-screened synthetic	.18
Addresses	Synthetic address grammar	.10
Emails	DNS-nonresolving domains	.08
Coined	Registry-screened synthetic	.03

Table 3. Field validation and retry comparison: GPT xhigh, 90 multi-field tasks $\times$ three trials per arm; $\pm$ gives 95% Wald margins. Task Pass@1 requires every field to be correct; field Pass@1 scores individual values.

Workflow	Task Pass@1	Field Pass@1
Batch submit; no field validation	.64 $\pm$.06	.78 $\pm$.03
Validate each field; retry errors	.82 $\pm$.05	.86 $\pm$.03

directed cells, Pass@1 is 67.0% on easy entities and 47.3% on hard ones; scaffolding raises these rates to 87.7% and 72.3%. Pooled Pass³ ranges from 3.3% for coined names to 46.7% for phone numbers and 60.0% for dates (Table 2). The pattern is consistent with format-constrained versus lexical capture. Dates and phone numbers follow tightly scoped numeric schemas. Medications reach 23.3% after unit normalization and come from a closed RxNorm vocabulary and coined names from a synthetic registry, but the agent is shown neither list.

Errors compound across fields. At three fields, only 54% of calls pass even though individual fields remain 75% accurate (Figure 5): one wrong field fails the entire call. Intermediate validation and retry counter this accumulation, raising task accuracy from 64% to 82% across the same multi-field tasks (Table 3). Because the intervention combines both mechanisms, we do not attribute the gain to either one alone.

Table 4. Regular agent-directed utterances with a clear speech-fidelity defect by system (lower is better). Scored n : 945, 1,142, 1,046, and 1,813, respectively.

	GPT minimal	GPT xhigh	Gemini high	Grok
Severity ≥ 2	2.5%	2.1%	3.6%	4.1%

Human review of 90 original reward-zero calls attributes 81 to the agent, one to user-simulator logic, four to infrastructure scoring errors, and leaves four unresolved; reported scores use the corrected grading. Among the 81 agent failures, agreed or adjudicated labels assign 42 as transcription, 16 as logical, six as VAD, and two as hallucination; the remaining 15 retain subtype disagreement. The speech-fidelity judge compares agent audio with its reference transcript, flagging mispronunciations, word additions, omissions, and clipping; severity ≥ 2 marks a clear,

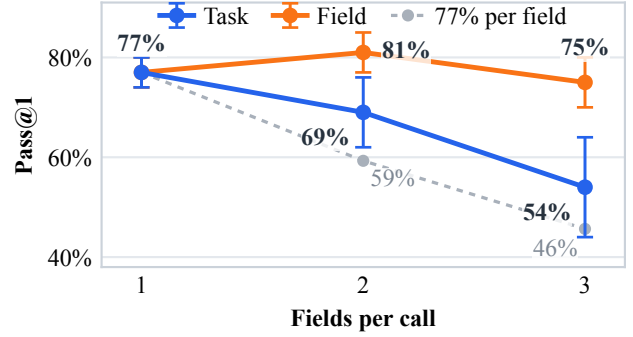


Fig. 5. Scaffolded GPT xhigh ($n = 630/180/90$ observations for one/two/three fields). The gray reference assumes independent 77% success per field. Bars show approximate 95% Wald intervals, not adjusted for repeated tasks; the single-field reference reuses matched calls.

potentially confusing mismatch (Table 4; findings overlapping the final second are excluded to prevent truncation false positives). The 60-utterance set yields .80 precision, .75 recall, and .77 F1. A separate GPT-5.5 xhigh text control passes all 200 tasks with the scaffolded policy, showing that the task set is straightforward for a capable text model.

6. CONCLUSION

Voice agents still struggle to collect exact entities reliably. Our caller is deliberately highly cooperative: it spells accurately and corrects faulty read-backs. Even under these favorable conditions, agent-directed systems achieve robust Pass³ on only 14–45% of tasks, compared with 37–58% with scaffolding.

Failures occur at both stages of recovery: agents do not consistently recognize risky captures, and when they do verify an error, only 30–37% are repaired.

Limitations

We study English, three providers, one simulator, and one run per main environment. We intentionally omit caller-side ASR to avoid confounding agent performance with a second recognizer. As a result, agent speech synthesis is not evaluated end to end: the speech-fidelity judge reports synthesis defects, but they do not yet affect task reward. The multi-field study is formative; broader provider, human-caller, and multilingual validation remain future work. Human validation of the fidelity judge covers Gemini and Grok only.

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7. COMPLIANCE WITH ETHICAL STANDARDS

The benchmark uses no real customer records or recordings. Values are synthetic or public and non-sensitive; calls use simulated callers and provider voices. Reviewers consented and were compensated.

Exact-value elicitation is dual use: it can improve service calls but can also facilitate collection of sensitive identifiers. Deployments should minimize collection, explain its purpose, protect stored values, and support human escalation. Our simulated English results are not evidence of safety, privacy, deployment readiness, or differences among people or demographic groups.

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